Well Ordering Principle

Every non-empty set S of non-negative integers contains a least element i.e. there is some integer 'a' in S $[\exists\, a \in S]$ such that $a \leq b$, $\forall\, b \in S$

Division Algorithm

[1 Theorem 3 Remarks]

Theorem[1]: Let 'a' and 'b' be two integers with $b > 0$.
Then there exists unique integers 'q' (quotient) and 'r' (remainder) such that $a = qb + r$, $[0 \leq r < b]$

Proof: Let $S = \{a - xb \mid x \text{ is an integer}, a - xb \geq 0\}$

① Claim: S is non-empty

① Infact $b > 0$, $b \in \mathbb{Z}$
$\therefore b \geq 1$
$\therefore |a|\, b \geq |a|$

② Taking $x = -|a|$, then
$a - xb = a - (-|a|\, b)$
$= a + |a|\, b$
$\geq a + |a|(1)$
≥ 0

③ Then $(a - (-|a|)\, b) \in S$ ($(a - xb) \in S$)
$\therefore S \neq \phi$
$\therefore$ S is non-empty

④ $\because a - xb \geq 0$
By well-ordering principle, S must ~~be~~ contain ~~the~~ a smallest element, say r

⑤ $\because r \in S$ and $r = a - qb$ for some integer q $\therefore$ $a = qb + r$, $r \geq 0$

② Claim: $r < b$

(# Proof by contradiction)

① assume $r \geq b$

② $\because r = a - qb$
Then $r - b = a - qb - b$
$= a - b(q + 1)$

③ $a - (q+1)b \in S$ ($\because r - b \geq 0$)
It is an element of S, since it is of the form a - (some integer) b

④ But $a - (q+1)b = r - b < r$

• $(r - b)$ is strictly smaller than r [$\because b > 0$] which is a contradiction to the fact that r is the least element of S (W.O.P)

⑤ $\therefore r \not\geq b$
$\Rightarrow r < b$

$\boxed{0 \leq r < b}$

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③ Claim: Uniqueness

① If possible, let $a = qb + r$, $0 \le r < b$

$a = q'b + r'$, $0 \le r' < b$

② Then $a = qb + r = q'b + r'$

$(r - r') = b(q' - q)$ $[\because b > 0]$

$\therefore |r - r'| = b|q' - q|$

③ Adding the two inequalities, $\begin{pmatrix} 0 \le r' < b \\ -b < -r' \le 0 \end{pmatrix}$

$-b < r - r' < b$

$\Rightarrow |r - r'| < b$

$\Rightarrow b|q' - q| < b$

$\Rightarrow |q' - q| < 1$

$\Downarrow$

implies $q' - q = 0$ $[\because q' - q$ is non-negative$]$

$\therefore q' = q$

$r' = r$

Hence, uniqueness

#Remark[1]: If a & b are integers, then $\exists$ q and r such that

$a = q'b + r'$, $b \le r' \le 2b$

Soln: Let $a = qb + r$, $0 \le r < b$ [DIVISION ALGORITHM]

$a = qb + r + b - b$

$a = (q - 1)b + (r + b)$ $\quad 0 \le r < b$

$a = q'b + r'$ $\quad b \le r' \le 2b$

[In general, $\boxed{a = q'b + r' \; ; \; nb \le r' < (n+1)b}$ for any integer n]

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#Remark 2: If a & b are integers with $b \neq 0$ then there exists a unique 'q' & 'r' ($\in \mathbb{Z}$) such that $a = qb + r$, $0 \leq r < |b|$

Soln: ① if $b < 0$, then $|b| > 0$ and $|b| = -b$ ← ask doubt

② $a = q|b| + r$, $0 \leq r < |b|$

$a = q(-b) + r$, $0 \leq r < |b|$

$a = (-q)b + r$, $0 \leq r < |b|$

#Remark 3: if a & b are integers with $b \neq 0$, then (Symmetric Remainder Form) prove that $\exists$ integers q and r s.t $a = qb + r$, $-\frac{1}{2}|b| < r \leq \frac{|b|}{2}$

Proof: ① for $b \neq 0$, We have $a = qb + r$, $0 \leq r < |b|$

② Divide interval $[0, |b|)$ into two parts: $\left[0, \frac{|b|}{2}\right]$ and $\left(\frac{|b|}{2}, |b|\right)$

③ If $r \in \left[0, \frac{|b|}{2}\right]$, $a = qb + r$, $-\frac{|b|}{2} < r \leq \frac{|b|}{2}$

④ If $r \in \left(\frac{|b|}{2}, |b|\right)$, then

$a = qb + r$

$\Rightarrow a = qb + r + |b| - |b|$

(i) if $b > 0$,

$|b| = b$

$\Rightarrow a = qb + r + b - b$

$= b(q+1) + (r-b)$

$= q'b + r'$ where $-\frac{|b|}{2} < r' < \frac{|b|}{2}$

(ii) if $b < 0$, then

$|b| = -b$

$a = qb + r - b + b$

$= b(q-1) + (r+b)$

$= q'b + r'$ where $-\frac{|b|}{2} < r' < \frac{|b|}{2}$.

$\therefore -\frac{|b|}{2} < r \leq \frac{|b|}{2}$

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Divisibility

The integer 'b' is said to be divisible by an integer $a \neq 0$ (non-zero integer) if there is an integer 'c' s.t $b = ac$

i.e $a \mid b \Leftrightarrow b = ac$ else $a \nmid b$

Properties: ① $a \mid 0$, $1 \mid a$, $a \mid a$ $\forall a$ (Universal Divisors) Imp: a should be non-zero

② $a \mid 1$, iff $a = \pm 1$ (Divisors of 1)

③ $a \mid b$ & $b \mid a \Rightarrow a = \pm b$ (Equality up to)

④ $a \mid b$ & $b \mid c \Rightarrow a \mid c$ (Transitive)

⑤ $a \mid b$ & $c \mid d \Rightarrow ac \mid bd$ (Multiplying Divisor)

⑥ $a \mid b$ & $a \mid c \Rightarrow a \mid bx \pm cy$ * for any $x, y \in \mathbb{Z}$ (Linear Combination)

☆ a divides b = b is a multiple of a
= a is a divisor of b

Greatest Common Divisor (GCD)

[2 Theorems]
[2 Corollarys, 1 Lemma]

Let a & $b \in \mathbb{Z}$, not both zero. Then the positive integer 'd' is gcd of a & b if : (i) $d \mid a$ & $d \mid b$
(ii) $c \mid a$ & $c \mid b$ for $c \leq d$
$[d = gcd(a, b)]$

Theorem 1: If a and b are integers, not both zero and $d = gcd(a, b)$, then $d = ax + by$ for some $x, y \in \mathbb{Z}$

Proof:
① Define $S = \{ax + by \mid ax + by > 0\}$
② Show S is non-empty [W.O.P]
③ Identify smallest element in S
④ Prove that d is common divisor of a & b (using D.A)
⑤ Prove $d = gcd(a, b)$

Let S be a set s.t $S = \{ax + by \mid ax + by > 0,\ x, y \in \mathbb{Z}\}$

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Claim: S is non-empty

① If $a \neq 0$, then $|a| = ax + by$, $x = \pm 1$, $y = 0$ $\therefore |a| \in S$ $S \neq \phi$

② By Well-Ordering Principle,
S must contain ~~least~~ a smallest element 'd'

③ Let d be the smallest element of S.

Claim: $d = \gcd(a, b)$

④ By division algorithm,

$$a = qd + r \quad 0 \leq r < d$$
$$\Rightarrow r = a - qd$$
$$\Rightarrow r = a - q(ax_0 + by_0)$$
$$r = a - qax_0 \mp qby_0$$
$$= a(1 - qx_0) + b(-qy_0)$$
$$= ax' + by'$$

⑤ This means $r \in S$ (by ④)
This contradicts that d is the least element in S as $0 \leq r < d$.
$\therefore \not\in \Rightarrow \boxed{r = 0}$

⑥ from ⑤, we can say $a = qd$ ($\because r = 0$)
$\therefore d \mid a$
Similarly $d \mid b$.

⑦ Let $c \mid a$ & $c \mid b$ then $c \mid ax_0 + by_0 = d$ (Divisibility L.C Principle)
$\therefore c \mid d$
$\therefore c \leq d$

$\therefore d = \gcd(a, b)$ proved

If $a, b \in \mathbb{Z}$, not both zero then $T = \{ax + by \mid x, y \in \mathbb{Z}\}$ is set of all multiples of $d = \gcd(a, b)$?

$\because d = \gcd(a, b) = ax_0 + by_0 \in T$
and $kd = akx_0 + bky_0 \in T$

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Theorem 2: If $\gcd(a,b)=1$ then a and b are said to be relatively prime

or

$d=\gcd(a,b)=1$, then $1=ax+by$, $x,y\in Z$

Proof $\Rightarrow$ we know $d=\gcd(a,b)$

Let $1=ax+by$ —①

Let $d=(a,b)$ —②

From ②, $d|a$ and $d|b$ —③

From ③, $d|ax+by=1$ (linear combination divisibility property)

$\therefore \boxed{d=1}$

Corollary1: if $d=\gcd(a,b)$ then $(a/d, b/d)=1$

Proof:

① $d=(a,b)$

② $d=ax+by$

③ $\therefore 1=\frac{a}{d}x+\frac{b}{d}y$ $(\div$ by $d)$ $\left[\frac{a}{d},\frac{b}{d}\in Z\right]$ $\because d|a$ & $d|b$

④ $\therefore (a/d, b/d)=1$

★ Properties:

① $\gcd(0,a)=|a|$ if $a\neq 0$

② $(a,-b)=(-a,b)=(-a,-b)=(a,b)$

If $a=qb+r$, p.t $(a,b)=(b,r)$

① Let $d=\gcd(a,b)$

$r=a-qb$

② Now $d|b$ & $d|r=a-qb$ $(\because d|a$ & $d|b)$

③ If $c|b$ & $c|r \Rightarrow c|a=qb+r$

Now $c|a$ & $c|b$ $\therefore c|d$ $(\because d=(a,b))$

④ $c\leq d$ or $d=\gcd(b,r)$

Corollary 2:

If $a|c$ & $b|c$ and $\gcd(a,b)=1$ then $ab|c$

① $\because a|c \Rightarrow$ we have $c=ar$, $r\in Z$

$b|c \Rightarrow$ we have $c=bs$, $s\in Z$

② $\because \gcd(a,b) \Rightarrow (ax+by=1)\times c$

$acx+bcy=1$

sub $c=ar=bs$

$\therefore a(bs)x+b(ar)y=c$

$\therefore c=ab(sx+cy)$

$\therefore \boxed{ab|c}$

proved

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EUCLID'S LEMMA [Imp]

If $a \mid bc$ with $\gcd(a,b)=1$
Then $a \mid c$.

Proof: ① $\gcd(a,b)=1$

② $\therefore ax + by = 1$
(multiply by c)
$acx + bcy = c$

③ $a \mid a \Rightarrow a \mid acx$
$a \mid bc \Rightarrow a \mid bcy$

④ $\therefore a \mid acx + bcy = c$
$\therefore \boxed{a \mid c}$

Eudidean Algorithm

$\Rightarrow$ to find $\gcd(a,b)$

Proof:
① Let there be $a,b \in \mathbb{Z}$ whose gcd is desired

② $\because \gcd(|a|,|b|) = \gcd(a,b)$
Let $a \geq b > 0$

③ Applying Division Algorithm,

$$a = q_1 b + r_1 \quad 0 \leq r_1 < b$$

If $r_1 = 0$, then $b \mid a$ and $\gcd(a,b) = b$

If $r_1 \neq 0$, then

④ $$b = q_2 r_1 + r_2 \quad 0 \leq r_2 < r_1$$

⑤ If $r_2 = 0$, $\gcd(b, r_1) = r_1 = \gcd(a,b)$ $[r_1 \mid b]$

$r_2 \neq 0$, then

$$r_1 = q_3 r_2 + r_3, \quad 0 \leq r_3 < r_2$$

⑤ Proceeding similarly until 0 remainder appears,

$$r_{n-2} = q_n r_{n-1} + r_n$$
$$r_{n-1} = q_{n+1} r_n + 0$$

dec. sequence of positive integers
$(b > r_1 > r_2 > r_3 \ldots > r_{n-1} \geq 0)$

Claim: $r_{j+2} < \frac{r_j}{2}$ (Euclidean Algo)

① In fact $r_{j+2} < r_{j+1} < r_j$

② If $r_{j+1} \leq \frac{1}{2} r_j$
Then $r_{j+2} < \frac{1}{2} r_j$

③ otherwise, $r_{j+1} > \frac{1}{2} r_j$

$r_j = 1 \times r_{j+1} + r_{j+2}$
$(\because 2r_{j+1} > r_j)$

$r_{j+2} = r_j - r_{j+1}$
$< \frac{1}{2} r_j \quad (\because r_{j+1} > \frac{1}{2} r_j)$

Lemma (already discussed)
If $a = qb + r$,
Then $\gcd(a,b) = (b,r)$
① $d = \gcd(a,b) \therefore d \mid a \;\&\; d \mid b$
② $r = a - qb$
③ $d \mid r = a - qb$ (from ①)
④ $d \mid r \;\&\; d \mid b$ (from ① & ③)
⑤ If $c \mid b \;\&\; c \mid r$ then $c = qb + r$ i.e $c \mid a$
⑥ $\because c \mid a \;\&\; c \mid b \Rightarrow \therefore c \mid d \;\therefore c \leq d, \therefore \gcd(b,r) = d$

⑥ $\therefore \gcd(a,b)$
$= \gcd(b, r_1)$
$= \gcd(b, r_2)$
$= \gcd(r_2, r_3)$
$\vdots$
$= \gcd(r_{n-1}, r_n)$
$= \gcd(r_n, 0)$
$= r_n$

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⊛ to find GCD eg: gcd(1547, 560) = ?

$1547x + 560y \Rightarrow x, y = ?$

Method 1 best for gcd

$1547 = 2 \times 560 + 427$

$560 = 1 \times 427 + 133$

$427 = 3 \times 133 + 28$

$133 = 4 \times 28 + 21$

$28 = 1 \times 21 + \boxed{7}$ ← last non-zero remainder

$21 = 3 \times 7 + 0$

$\therefore \gcd(1547, 560) = 7$

① first find gcd using Euclidean Algo

② Then back track L.C of $\overset{d=}{7}$ back to a & b to get x & y.

$7 = 28 - 1 \times 21$

$= 28 - 1(133 - 4 \times 28)$

$= 5 \times 28 - 1(133)$

$= 5 \times 28 (427 - 3 \times 133) - 1(133)$

$= 5 \times 427 - 16(133)$

$= 5 \times 427 - 16(560 - 427)$

$= 21 \times 427 - 16(560)$

$= 21(1547 - 2 \times 560) - 16(560$

$= 21(1547) - 58(560)$

$\therefore 7 = 1547x + 560y$

$x = 21, y = -58$

Method 2:

$1547x_i + 560y_i = r_i$

① Start by expressing a & b as r by multiplying 0 & 1

$\Rightarrow 1547(1) + 560(0) = 1547$ — ①

$\Rightarrow 1547(0) + 560(1) = 560$ — ②

② divide r_1 & r_2

$\frac{1547}{560} = 2.7625$

③ take integer part multiply it into the 2nd eqn & subtract to get 3rd new equation

④ repeat the process

(2×) ⊖

$1547(1) + 560(-2) = 427$ — ③ ; $\frac{560}{427} = 1\ldots$

(1×) ②⊖③

$1547(-1) + 560(3) = 133$ — ④ ; $\frac{427}{133} = 3\ldots$

(3×) ③−④

$1547(4) + 560(-11) = 28$ — ⑤ ; $\frac{133}{28} = 4$

(4×)

$1547(-17) + 560(47) = 21$; $\frac{28}{21} = 1\ldots$

(1×) −

$1547(21) + 560(-58) = 7$; $\frac{21}{7} = 3$

(3×)

$1547(\ ) + 560(\ ) = 0$

$\gcd = 7 = 1547(21) + 560(-58)$

$x = 21$

$y = -58$

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Method 3: (best) for L.C & gcd

i	q_{i+1}	r_i	x_i	y_i
-1		$a = 1547$	1	0
0	2	$b = 560$	0	1
1	1	$r_1 = 427$	1	-2
2	3	133	-1	3
3	4	28	4	-11
4	1	21	-17	47
5	3	7	21	-58
		0		

(multiply → q; subtract)

① divide $\frac{a}{b}$ and get q_1

② fill q in q column and multiply with x_i & y_i

③ subtract consecutive x_i's and y_i's till u reach rem = 0

Gaussian Integers

Complex numbers where real, imag part are integers

$\{\alpha = m+ni \ , \ m, n \in \mathbb{Z} \} = \mathbb{Z}[i]$

We say that $\alpha \mid \beta$ if there is a $\gamma = \mathbb{Z}[i]$ s.t $\beta = \alpha\gamma$

[α, β, γ are Gaussian integers]

gcd $(\alpha+i\beta, a+ib) = m+in$, if Gaussian integer $m+in$ divides both $\alpha+i\beta, a+ib$ with highest magnitude.

#Note ⇒ gcd of Gaussian integers is not unique

If $m+ni$ is gcd then $\pm 1\,(m+ni)$ & $\pm i\,(m+ni)$ are also gcd

eg: find gcd $(5+6i, 3-2i)$

① find absolute value i.e $|5+6i| = \sqrt{5^2+6^2} > |3-2i| = \sqrt{3^2+2}$

② $\therefore \frac{\text{bigger}}{\text{smaller}} \Rightarrow \frac{5+6i}{3-2i} = \frac{(5+6i)(3+2i)}{(3-2i)(3-2i)} = \frac{15+28i-12}{13} = \frac{3+28i}{13}$

$= \frac{3}{13} + \frac{28}{13} i$

$\approx = 0+2i$ [nearest Gaussian integer]

③ Euclidean Algo

$(5+6i) = 2i\,(3-2i) + 1$ (4+6i)

$(3-2i) = 1\,(3-2i) + 0$

$\therefore$ gcd = last non-zero remainder $= (5+6i, 3-2i) = 1$

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Prime Numbers as sum of 2 squares

Let $p \mid b^6+1$ where $b \in \mathbb{Z}$, p is a prime

Then express $p = c^2+d^2$ (sum of 2 squares)

$$\left[\begin{array}{l} p = c^2+d^2 \\ \quad = (c+di)(c-di) \end{array}\right\} \begin{array}{l} c+di \mid p+0i \\ \text{or } c-di \mid p+0i \end{array} \Bigg] \text{ Basic explanation}$$

$(b^6+1) = (b^2+1)(b^4-b^2+1)$

then either: $p \mid b^2+1$ or $p \mid b^4-b^2+1$ (as p is a prime, $p \mid ab \Rightarrow p \mid a$ or $p \mid b$)

STEPS:

(i) Let $p \mid b^2+1 = (b+i)(b-i)$

$\Rightarrow$ Find gcd of $(p; b+i) = (c+di)$ $\left[\begin{array}{c} p = c^2+d^2 = (c+di)(c-di) \\ c+di \mid p \Leftrightarrow c-di \mid p \end{array}\right]$

(ii) Let $p \mid b^4-b^2+1$

$$b^4-b^2+1 = (b^2-1)^2 + b^2$$
$$= \left[(b^2-1)^2 + bi\right]\left[(b^2-1) - bi\right]$$

$\Rightarrow$ Fund gcd $(p, (b^2-1)+bi) = (c+di)$

$\therefore p = c^2+d^2$

Least Common Multiple

For 2 integers 'a' & 'b', $\Rightarrow$ lcm $(a,b) = [a,b]$ is the least possible integer 'm' such that (i) $a \mid m$ & $b \mid m$

(ii) if $a \mid c$ & $b \mid c$ for some $c > 0$ then $m \leq c$

Note : $\gcd(a,b) \times \text{lcm}[a,b] = |ab|$

$$[a,b] = \frac{|ab|}{(a,b)}$$

Proof : $d = \gcd(a,b)$

$d \mid a$ & $d \mid b$

Let $a = dr$, $b = ds$ $(r,s \in \mathbb{Z})$

If $m = \frac{ab}{d}$ then $m = as = rb$ $\left(\because m = \frac{ab}{d} = \frac{(dr)b}{d} = rb = \left(\frac{a}{d}s\right) = as\right)$ $\therefore$ m is common multiple for a, b

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CONGRUENCE THEORY

$a \equiv b \pmod m \iff m \mid a-b$ (m divides a−b)

$\iff a = km + b$ (b is the remainder when m divides a)

eg: $8 \equiv 3 \pmod 5 \Rightarrow 5 \mid 8-3 \Rightarrow 5 \mid 5$ or $8 = 1(5)+3$

$8 \equiv -2 \pmod 5 \Rightarrow 5 \mid 8-(-2) \Rightarrow 5 \mid 10$ or $8 = 2(5)+(-2)$

Properties: (equivalence relation)

① $a \equiv a \pmod m$ $\quad m \mid a-a \Rightarrow m \mid 0$ (reflexive)

② $a \equiv b \pmod m \iff b \equiv a \pmod m$ (symmetric)

③ $a \equiv b \pmod m$, $b \equiv c \pmod m \Rightarrow a \equiv c \pmod m$ (transitive / associative)

$(a-b \mid m \ \& \ b-c \mid m, \ a-c = (a-b)+(b-c) = mh+mk = m(h+k))$

✱ ④ $a \equiv b \pmod m$, $c \equiv d \pmod m \Rightarrow a \pm c \equiv (b \pm d) \pmod m$, $ac \equiv bd \pmod m$

✱ ⑤ $a \equiv b \pmod m \Rightarrow a \pm c \equiv b \pm c \pmod m$, $ac \equiv bc \pmod m$

✱ ⑥ $a \equiv b \pmod m \Rightarrow a^k \equiv b^k \pmod m$ $(k>0)$ [converse not true]

✱ ⑦ $a \equiv b \pmod m$, $a \equiv b \pmod n \Rightarrow a \equiv b \pmod{mn}$ if $\gcd(m,n)=1$ (coprime)

Note: $a^k \equiv b^k \pmod m \not\Rightarrow a \equiv b \pmod m$ [$5^2 \equiv 4^2 \pmod 3$ ✓ but $5 \equiv 4 \pmod 3$ ×]

Theorem 1: if $a \equiv b \pmod n$,

prove that: $\gcd(a,n) \equiv \gcd(b,n)$

① $n \mid a-b$ $(\because a \equiv b \pmod n)$

② $\therefore (a-b) = kn$ for some $k \in \mathbb{Z}$

③ Let $d_1 = \gcd(a,n)$ $\therefore d_1 \mid a$ & $d_1 \mid n$

$d_2 = \gcd(b,n)$ $d_2 \mid b$ & $d_2 \mid n$

$\Rightarrow a = d_1 r$, $n = d_1 s$ } when $(r,s) = 1$

④ $b = a - kn$

sub from ③

$b = d_1 r - k d_1 s$

$b = d_1 (r - ks)$

$\therefore d_1 \mid b$

⑤ $\because d_1 \mid b$ & $d_1 \mid n$

$\therefore d_1 \mid \gcd(b,n)$

⑥ $\therefore d_1 \mid d_2 \Rightarrow d_1 \leq d_2$

⑦ Similarly we can prove that $d_2 \leq d_1$

⑧ from ⑥ & ⑦

$d_1 = d_2$ or

$\gcd(a,n) = \gcd(b,n)$

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Theorem 2: If $ca = cb \pmod n$

Then $a = b \pmod{\frac{n}{d}}$ where $d = \gcd(c, n)$

Proof:

① $n \mid ca - cb$

② $ca - cb = kn$

③ Let $d = \gcd(c, n)$

Then $d \mid c$ & $d \mid n$

Let $c = dr$ & $n = ds$, $\gcd(r, s) = 1$

④ (from ②, ③)

$ca - cb = kn$

(sub from ③)

$dr(a - b) = k(ds)$

⑤ $ks = r(a - b) \quad \therefore s \mid (a - b)r$

⑥ $\because \gcd(r, s) = 1$, Euclid's Lemma

$s \mid a - b$

⑦ $a \equiv b \pmod s$ from ⑥

$\boxed{a \equiv b \pmod{\frac{n}{d}}}$ from ③

proved

Corollary

if $ca \equiv cb \pmod n$

Then $a \equiv b \pmod n$

if $\gcd(c, n) = 1$

eg: $8 \times 3 = 3 \times 3 \pmod 5$

$8 \equiv 3 \pmod 5$ as $(3, 5) = 1$

Equivalence Class and Inverses.

Equivalence class $\bar{a}$ = {set of integers which leave remainder a, when divided by m}

$\{\bar{0}, \bar{1}, \bar{2}, \ldots \overline{m-1}\} \Rightarrow$ complete set of residues (mod m)

$= \mathbb{Z}m$

eg: $\mathbb{Z}_5 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ Here $\bar{2}$ means {all integers that give remainder 2, when divided by 5}

Note: Consider $\mathbb{Z}_n$ with multiplication$_n$ and addition$_n \pmod n$

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Claim: $a \in \mathbb{Z}_n$ has a multiplicative inverse b if $(a,n)=1$

$a^{-1} \pmod m = b$ iff $(a,m)=1$

① Let $d = \gcd(a,n) > 1$

② Let $b \in \mathbb{Z}_n$ such that $ab \equiv 1 \pmod m$

$\therefore n \mid ab-1$

③ $\because n \mid ab-1$, $\therefore d \mid ab-1$ (as $d \mid a$ & $d \mid n$)

But $d \mid a$ already (from ①)

$\therefore d \mid ab$

④ $\Rightarrow \therefore d \mid (ab-1) - ab$

$d \mid (-1)$

$d = 1$, which is a CONTRADICTION ($\because d>1$ assumed)

⑤ On the other hand, let $d = (a,n) = 1$

$ax + ny = 1$ [for some $x, y \in \mathbb{Z}$]

$ax \equiv 1 \pmod n$ [$\because n \equiv 0 \bmod n$]

$\Rightarrow x$ is the multiplicative inverse of $a \pmod n$

⊛ to find inverse

① check for gcd

② if $\gcd \neq 1$, inverse does not exist

if $\gcd = 1$, find $ax+by = 1$ ← gcd

③ x is the inverse of 'a'

Note: In $\mathbb{Z}_p = \{0, 1, 2, 3 \ldots p-1\} \pmod p$ $\forall a \in \mathbb{Z}_p$, $a \neq 0$

$a^{-1} \pmod p$ exists [$\because \gcd(a,p) = 1$, $p \to$ prime]

① Consider the congruence $ax \equiv b \pmod m$ with $0 < a, b < m$

② If $\gcd(a,m) = 1$, then congruence has a unique soln, say $x_0 = a^{-1} b \pmod m$

(other solns are $x_0 + km$, $\forall k \in \mathbb{Z}$)

③ Moreover $ax \equiv b \pmod m$ has a soln iff $g.c.d(a,m) = d \mid b$

④ Here $ax \equiv b \pmod m \Rightarrow \left(\frac{a}{d}\right)x \equiv \left(\frac{b}{d}\right) \bmod \left(\frac{m}{d}\right)$

i.e $a'x = b' \pmod{m'}$

has a unique soln say $x_0 = (a')^{-1} b' \pmod{m'}$

⑤ There are 'd' incongruent solns

x_0', $x_0' + m'$, $x_0' + 2m'$ ... $x_0' + (d-1)m'$

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Steps to solve linear congruences

$$a x \equiv b \pmod{m}$$

$a, b, m \in \mathbb{Z}$, $m > 0$

① find $\gcd(a, m) = d$

② If $d \nmid b \Rightarrow$ no soln

If $d \mid b \Rightarrow$ soln exists

③ $\frac{a}{d} x = \frac{b}{d} \pmod{\frac{m}{d}}$

$\therefore \gcd\left(\frac{a}{d}, \frac{m}{d}\right) = 1$

④ find modular inverse of $a' \pmod{m'}$ such that

$a' \cdot (a')^{-1} \equiv 1 \pmod{m'}$

(can be found using $ax + by = d$, $x \rightarrow$ if a^{-1} needed.)

⑤ multiply by inverse on both sides

⑥ Write all solution

$x = x_0 + k m' \quad (k = 0, 1, 2 \ldots d-1)$

(if $d = 1$, only one soln exists)

Theorem: If $a \equiv b \pmod{n_1}$ and $a \equiv b \pmod{n_2}$, then prove that $a \equiv b \pmod{n}$ where $n = \text{lcm}(n_1, n_2) = [n_1, n_2]$

Proof: ① we know, $\gcd(n_1, n_2) * \text{lcm}[n_1, n_2] = n_1 \cdot n_2$

② $a \equiv b \pmod{n_1} \Rightarrow a - b = k_1 n_1$

$a \equiv b \pmod{n_2} \Rightarrow a - b = k_2 n_2$

$\} \; k_1, k_2 \in \mathbb{Z}$

③ Let d be $\gcd$ of (n_1, n_2)

$\therefore n_1 = dr$, $n_2 = ds$, $\gcd(r, s) = 1$ ⋮ $\frac{n_1}{dr} = 1$

④ $a - b = k_2 n_2 = k_2 n_2 \frac{n_1}{dr}$

$= \frac{k_2}{r} \frac{n_1 \cdot n_2}{d}$

$= \frac{k_2}{r} \times \text{lcm}(n_1, n_2)$

$a - b = \frac{k_2}{r} \times n$

⑤ $\therefore n \mid a - b$ if k_2/r is an integer

⑥ to prove that $r \mid k_2$

$a - b = k_1 n_1 = k_2 n_2$

$k_1(dr) = k_2(ds)$

$k_1 r = k_2 s$

$\therefore r \mid k_2 s$

$\therefore r \mid k_2 \; (\because \gcd(r, s) = 1)$

⑦ $\therefore$ ~~a b a k n~~ or $n \mid a - b$

or $\boxed{a \equiv b \pmod{n}}$

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#Note: If $ab \equiv cd \pmod n$

and $b \equiv d \pmod n$ with $\gcd(b,n)=1$

Then $a \equiv c \pmod n$

Proof: ① $b \equiv d \pmod n$

② $cb \equiv cd \pmod n$ [property of congruence]

③ $ab \equiv cd \pmod n \Leftrightarrow cd \equiv ab \pmod n$

④ from ② & ③, $ab \equiv cb \pmod n$ [associativity]

$\therefore \boxed{a \equiv c \pmod n}$ $\because \gcd(b,n)=1$

Chinese Remainder Theorem

Consider
$$\left.\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_r \pmod{m_r} \end{aligned}\right\} \rightarrow \text{system of } {}^{c}r^{a} \text{ congruences}$$

Let $\gcd(m_i, m_j)=1$ for $i \neq j$ for all $i, j = 1, 2, 3 \ldots r$

The system has a unique solution $\Rightarrow \boxed{x \pmod M}$ where $M = m_1 . m_2 . m_3 \ldots m_r$

Existence of solution

$M_i = \frac{M}{m_i}$ $\forall i$ i.e $M_i = m_1 . m_2 \ldots m_{i-1} m_{i+1} \ldots m_r$

$\Rightarrow (M_i, m_i) = 1$

$\Rightarrow M_i^{-1} \pmod{m_i} = N_i$ (exists)

$M_i N_i \equiv 1 \pmod{m_i}$ $\forall i$

Let $x = \sum_{i=1}^{r} a_i M_i N_i = a_1 M_1 N_1 + a_2 M_2 N_2 + \ldots a_r M_r N_r$

This x satisfies all congruences i.e $x \equiv a_i \pmod{m_i}$

Uniqueness

. If possible, let x, y satisfy the system

$x \equiv a_i \pmod{m_i}$ $\forall i$
$y \equiv a_i \pmod{m_i}$

$\Rightarrow x - y \equiv 0 \pmod{m_i}$
$x \equiv y \pmod{m_i}$
$x \equiv y \pmod{m_1 m_2 \ldots m_r}$ [$\because (m_i, m_j)=1$ if $i \neq j$]

$\therefore x \equiv y \pmod M$

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eg: $2x \equiv 1 \pmod 5$
$3x \equiv 9 \pmod 6$ solve for x
$4x \equiv 1 \pmod 7$
$5x \equiv 9 \pmod{11}$

$2x \equiv 1 \pmod 5 \Rightarrow x \equiv 3 \pmod 5$ ($\because 2x-1 \mid 5$ for $x=3$)

$3x \equiv 9 \pmod 6 \Rightarrow x \equiv 3 \pmod 2 \Rightarrow x \equiv 1 \pmod 2$ (÷ by 3 first : common multiples)

$4x \equiv 1 \pmod 7 \Rightarrow x \equiv 2 \pmod 7$ ($\because 4x-1 \mid 7$ for $x=2$)

$5x \equiv 9 \pmod{11} \Rightarrow x \equiv 4 \pmod{11}$ ($\because 5x-9 \mid 11$ for $x=4$, $20-9=11 \mid 11$)

$a_i \quad m_i$

$x \equiv 3 \pmod 5$
$x \equiv 1 \pmod 2$
$x \equiv 2 \pmod 7$
$x \equiv 4 \pmod{11}$

$M = 5 \times 2 \times 7 \times 11 = 770$

$M_1 = \frac{770}{5} = 154$ $\quad N_1 \equiv (154)^{-1} \pmod 5 \equiv (4)^{-1} \bmod 5 \equiv 4$ [(4 × something − 1) | 5]

$M_2 = \frac{770}{2} = 385$ $\quad N_2 = (385)^{-1} \pmod 2 \equiv (1^{-1}) \bmod 2 = 1$

$M_3 = \frac{770}{7} = 110$ $\quad N_3 = (110)^{-1} \pmod 7 \equiv (5^{-1}) \bmod 7 = 3$ [(5 × something^3 − 1) | 7]

$M_4 = \frac{770}{11} = 70$ $\quad N4 = (70^{-1}) \pmod{11} = (4^{-1}) \bmod 11 = 3$

Note: $y^{-1} \pmod m$ can be thought as $y * x = 1 \pmod m$ ← need to find out

eg: $(154^{-1}) \bmod 5$

$154x = 1 \bmod 5$.
$4x = 1 \pmod 5$
$x = 4 \pmod 5$

$\therefore x = \sum_{i=1}^{4} a_i M_i N_i = 3(154)(4) = 1848 = 3733 \pmod{770}$
$+ 1(385)(1) \quad + 385$
$+ 2(110)(3) \quad + 660$
$+ 4(70)(3) \quad + 840$
$= \boxed{653 \pmod{770}}$

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EULER PHI FUNCTION (φ)

$$\varphi(n) = \left| \left\{ b \in \mathbb{Z} \mid \begin{matrix} \gcd(b,n)=1 \\ 0 \leq b \leq n \end{matrix} \right\} \right|$$

↑ number of +ve integers less than n and relatively prime with n (coprime)

$\varphi(1) = 1$
$\varphi(2) = 1$
$\varphi(3) = 2$

(*) ① $\boxed{\varphi(p) = p-1}$, $p \to$ prime

② $\boxed{\varphi(p^\alpha) = p^\alpha - p^{\alpha-1} = p^\alpha\left(1 - \frac{1}{p}\right)}$

Theorem: Euler phi function is multiplicative
i.e $\varphi(mn) = \varphi(m)\,\varphi(n)$, provided $\gcd(m,n)=1$ [coprime]

Proof:

① Let $S = \left\{ i \in \mathbb{Z} \mid \begin{matrix} 0 \leq i \leq mn \\ (i, mn) = 1 \end{matrix} \right\} \Rightarrow |S| = \varphi(mn)$

$S_1 = \left\{ i \in \mathbb{Z} \mid \begin{matrix} 0 \leq i \leq m \\ (i, m) = 1 \end{matrix} \right\} \Rightarrow |S_1| = \varphi(m)$

$S_2 = \left\{ i \in \mathbb{Z} \mid \begin{matrix} 0 \leq i \leq n \\ (i, n) = 1 \end{matrix} \right\} \Rightarrow |S_2| = \varphi(n)$

② Consider the congruences $\left\{ \begin{matrix} j \equiv j_1 \pmod m \\ j \equiv j_2 \pmod n \end{matrix} \right\}$

③ As $(m,n) = 1$, by Chinese Remainder Theorem } for every pair (j_1, j_2) with $(j_1, m) = (j_2, n) = 1$,

There is a unique j such that $(j, mn) = 1$

i.e no of pairs $(j_1, j_2) \in S_1 \times S_2$ = no. of $j \in S$

④ By counting principle, $|S| = |S_1|\,|S_2|$

$\boxed{\varphi(mn) = \varphi(m)\,\varphi(n)}$

proved.

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⊛ If $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_r^{\alpha_r}$

then $\varphi(n) = n \cdot \prod_{p \mid n} \left(1 - \frac{1}{p}\right)$

#Note: for a natural number n, which is a product of 2 primes p & q, the knowledge of $\varphi(n)$ is same as knowledge of primes p & q.

Proof:

① $n = pq$

② $\varphi(n) = \varphi(pq)$

$= \varphi(p)\,\varphi(q)$

$= (p-1)(q-1)$

③ If n is even, then $p = 2$ and $q = \frac{n}{2}$

$\Rightarrow \boxed{\varphi(n)} = (2-1)\left(\frac{n}{2} - 1\right) = \boxed{\frac{n}{2} - 1}$

④ If n is odd,

(i) if p, q known $\Rightarrow \varphi(n) = (p-1)(q-1)$

$= pq - (p+q) + 1$

$= (n+1) - (p+q)$

⑤ If $\varphi(n)$ is known and $\varphi(n)$ = even

$\varphi(n) = (n+1) - (p+q)$

even — even $\Rightarrow$ p = odd, q = odd

⑥ $\therefore p+q = (n+1) - \varphi(n)$

even − even

= say, $2b$

⑦ $p+q = 2b$, $pq = n$ } p, q roots of $x^2 - (p+q)x + pq = 0$

$x^2 - 2bx + n = 0$

$\therefore$ ⑧ $x = \frac{2b \pm \sqrt{4b^2 - 4n}}{2} = \boxed{b \pm \sqrt{b^2 - n}}$

$b = \frac{p+q}{}$

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DIOPHANTINE EQUATION $(ax+by=c)$ $[a,b,c \in \mathbb{Z}]$

Theorem: The linear Diophantine Equation
$ax+by=c$ has a solution if and only if
$d \mid c$, where $d=\gcd(a,b)$

Particular soln $\Rightarrow x_0, y_0$

Other solns $\Rightarrow x = x_0 + (b/d)t \quad \forall\, t \in \mathbb{Z}$
$y = y_0 - (a/d)t \quad d=\gcd(a,b)$

⊛ **Theorem:**

"The system of linear congruences
$ax+by \equiv r \pmod n$
$cx+dy \equiv s \pmod n$,
has a unique soln modulo n whenever
$\gcd(ad-bc, n)=1$ "

Proof:

① $(ax+by \equiv r \pmod n) \times d$
$adx+bdy \equiv rd \pmod n$ —— ⓐ

② $(cx+dy \equiv s \pmod n) \times b$
$bcx+bdy \equiv bs \pmod n$ —ⓑ

③ ⓐ − ⓑ
$(ad-bc)x \equiv (rd-bs) \pmod n$ —ⓒ

④ Let $\gcd(ad-bc, n)=1$
Then $(ad-bc)z \equiv 1 \pmod n$ possesses a unique soln 't'

⑤ multiply t with ⓒ to get
$x \equiv t(dr-bs) \pmod n$

⑥ similarly get y.
$y \equiv t(as-cr) \pmod n$

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PRIMES

Prime ⇒ an integer $p>1$, if its only +ve divisors are 1 & p

[1 → neither prime nor composite]

✱ Theorem : If p is prime and $p|ab$, then $p|a$ or $p|b$

Proof : ① If $p|a$, then no need to prove

② Assume $p \nmid a$

⇒ $\gcd(p,a)=1$ (∵ divisors of p are p & 1)

⇒ ∴ $p|b$ (by Euclid's Lemma)

[If $a|bc$, with $\gcd(a,b)=1$
Then $a|c$]

#. Note : ① If p is a prime and $p|a_1 a_2 \cdots a_n$,
Then $p|a_i$ for some $1 \le i \le n$

② If $p, q_1, q_2, \ldots q_n$ are all primes,
and $p|q_1 q_2 q_3 \ldots q_n$,
Then $p = q_i$ for some $1 \le i \le n$

Fundamental Theorem of Arithmetic

Any natural number can be expressed uniquely as a product of primes (uniqueness is upto order of primes)

$$n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot p_3^{\alpha_3} \cdots p_n^{\alpha_n}$$

Proof : ① If n is prime, then $n=n$ is the equivalent representation.

② If n is composite, then n has a ~~composite~~ non-trivial divisor say a
⇒ $a|n$ or $n=ab$ for some $b \in \mathbb{Z}$

③ Now, $n=ab$ & $1<a<n$, $1<b<n$

It is enough if we prove theorem for a & b instead of n

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④ Repeating the same argument for a, b and smaller factors, all greater than 1, less than n, we get primes $p_1, p_2 \ldots, p_k$ such that:

$$n = p_1 \cdot p_2 \cdot p_3 \cdots p_k$$

✱ Uniqueness

① Let $n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ and

$n = q_1^{\beta_1} \cdot q_2^{\beta_2} \cdots q_r^{\beta_r}$ be two distinct representations of n.

② Then, $p_i \mid n = q_1^{\beta_1} \cdots q_r^{\beta_r} \ \forall i$

$\Rightarrow p_i = q_j$ for some j, $1 \leq j \leq r$

③ Now, $q_j^{\beta_j} = p_i^{\beta_j}$ and $p_i^{\alpha_i} \mid n$

$\Rightarrow p_i^{\alpha_i} \mid p_i^{\beta_j}$

$\Rightarrow \alpha_i \leq \beta_j$

For similar argument, $\beta_j \leq \alpha_i$

$\Rightarrow \alpha_i = \beta_j$ whenever $p_i = q_j$

$\Rightarrow r = k$

Canonical Representation

For $n > 1$, the representation $n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdots p_r^{\alpha_r}$ with $p_1 < p_2 < p_3 \cdots < p_r$ is a canonical representation of 'n as a product of prime numbers.'

eg: $4200 = 2^3 \times 3 \times 5^2 \times 7$

GCD ⇒ If $a = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$

$b = p_1^{\beta_1} \cdots p_r^{\beta_r}$

Then, $\gcd(a, b) = p_1^{\gamma_1} \cdots p_r^{\gamma_r}$

where each $\gamma_i = \min\{\alpha_i, \beta_i\}$

$p^{\alpha} \| a$ (i.e p^{α} exactly divides a) if $p^{\alpha} \mid a$ & $p^{\alpha+1} \nmid a$

[i.e α is the largest exponent of p, such that $p^{\alpha} \mid a$]

Note: ① If $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$ (canonical representation)

Then $p_i^{\alpha_i} \| n$

② If $n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdots p_r^{\alpha_r}$, then, no. of divisors of n is equal to $(\alpha_1 + 1)(\alpha_2 + 1) \cdots (\alpha_r + 1)$

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FERMAT'S LITTLE THEOREM

If p is a prime and $p \nmid a$, then: $a^{p-1} \equiv 1 \pmod{p}$

Proof:

① The set $\{0 \cdot a, 1 \cdot a, 2 \cdot a, \ldots, (p-1)a\}$
is a complete set of residues $(\text{mod } p)$

② $(\because ia \equiv ja \pmod{p}$ for some $1 \leq i, j \leq p-1)$

$p \mid ia - ja$

$p \mid (i-j)a$

$\therefore p \mid (i-j)$ [as $p \nmid a$]

$\therefore i - j = 0 \quad (\because i, j < p)$

$i = j$

Hence, the set is a complete set of residues modulo p

③ i.e the sets $\{0, 1.a, 2a, \ldots (p-1)a\}$ and $\{0, 1, \ldots p-1\}$ are simply a rearrangement and are same when considered $(\text{mod } p.)$.

④ $a.(2a).(3a) \ldots (p-1).a \equiv 1 \times 2 \times 3 \times \ldots (p-1) \bmod p$

$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$

$\therefore p \mid (p-1)! [a^{p-1} - 1]$

$p \mid (p-1)!$ or $p \mid a^{p-1} - 1$

(does not divide)

$\therefore p \mid a^{p-1} - 1 \quad (\text{as } p \nmid (p-1)!)$

$$\boxed{a^{p-1} \equiv 1 \pmod{p}}$$

or $$\boxed{a^p \equiv a \pmod{p}}$$

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$$(a+1)^p = a^p + \binom{p}{1}a^{p-1} + \dots \binom{p}{p-1}a + 1 \quad \text{[Binomial Theorem]}$$

$\binom{p}{p-1}$ ⟵ ①

$\binom{p}{k} \equiv 0 \pmod p$ for $1 \le k \le p-1$

$$\left[\because ({}^pC_k) = \binom{p}{k} = \frac{p!}{k!(p-k)!}\right]$$

$$k!\binom{p}{k} \equiv \frac{p!}{(p-k)!}$$

$$k!\binom{p}{k} \equiv p(p-1)(p-2)\dots(p-k+1) \equiv 0 \pmod p$$

$$p \mid k!\binom{p}{k}$$

$$\therefore p \mid \binom{p}{k} \quad [\because p \nmid k! \;\; 1 \le k < p]$$

$$\boxed{\binom{p}{k} \equiv 0 \pmod p} \quad ②$$

$\therefore$ From ① & ②, # Note $\Rightarrow$ $(a+1)^p \equiv (a^p+1) \pmod p$

Note $\Rightarrow$ $(a+b)^p \equiv a^p + b^p \pmod p$

$(a+b)^{p^2} \equiv a^{p^2} + b^{p^2} \pmod p$

$(a+b)^{p^k} \equiv a^{p^k} + b^{p^k} \pmod p$

Note: if $p \mid a^n \Rightarrow p^n \mid a^n$

(Continue more from LMS FLT pg 4 onwards)

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FLT Continued (Pg 5)

Lemma: If p and q are distinct primes

with $\left.\begin{aligned} a^p &\equiv a \pmod q \\ a^q &\equiv a \pmod p \end{aligned}\right] \Rightarrow$ then $a^{pq} \equiv a \pmod{pq}$

Proof:

① $a^{p-1} \equiv 1 \pmod p$ [Fermat's Little Theorem]

$\Rightarrow a^p \equiv a \pmod p$ ——— ⊛

② $a^q \equiv a \pmod p$ given

$\therefore (a^q)^p \equiv a^p \pmod p$

$\equiv a \pmod p$ — (from ⊛)

$\therefore a^{pq} \equiv a \pmod p$

③ $a^p \equiv a \pmod q$

$(a^{qp})^q \equiv a^q \pmod q$

$\equiv a \pmod{pq}$ (⊛)

$\therefore a^{pq} \equiv a \pmod q$

④ from ② & ③, $\left.\begin{aligned} a^{pq} &\equiv a \pmod p \\ a^{pq} &\equiv a \pmod q \end{aligned}\right] \therefore \boxed{a^{pq} \equiv a \pmod{pq}}$

[$\because \gcd(p,q) = 1$
p, q are primes]

If p is odd prime, prove that: $1^{p-1} + 2^{p-1} + 3^{p-1} + \ldots (p-1)^{p-1} \equiv -1 \pmod p$

① $a^{p-1} \equiv 1 \pmod p$ [FLT]

② $\therefore 1^{p-1} \equiv 1 \pmod p$

$2^{p-1} \equiv 1 \pmod p$

$\vdots$

$(p-1)^{p-1} \equiv 1 \pmod p$

③ adding ②,

$1^{p-1} + 2^{p-1} + \ldots (p-1)^{p-1} \equiv (p-1) \pmod p$

$\equiv -1 \pmod p$

proved

If p is an odd prime, prove that: $1^p + 2^p + 3^p + \ldots (p-1)^p \equiv 0 \pmod p$

① $a^{p-1} \equiv 1 \pmod p$ [FLT]

$\therefore a^p \equiv a \pmod p$

② $1^p \equiv 1 \pmod p$

$2^p \equiv 2 \pmod p$

$\vdots$

$(p-1)^p \equiv p \pmod p$

③ Adding eqns of 2

$1^p + 2^p + \ldots (p-1)^p \equiv 1 + 2 + \ldots (p-1) \pmod p$

$\equiv \frac{p(p-1)}{2} p \pmod p$

$1^p + 2^p + \ldots (p-1)^p \equiv 0 \pmod p$

$\because \frac{p-1}{2}$ is an integer.

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Two odd primes p & q, such that $p-1 \mid q-1$. If gcd $(a, pq) = 1$ then show that: $a^{q-1} \equiv 1 \pmod{pq}$

① $(a, pq) = 1$
$\therefore (a, p) = (a, q) = 1$

② $a^{p-1} \equiv 1 \pmod{p}$
$a^{q-1} \equiv 1 \pmod{q}$ [Using FLT]

③ $a^{k(p-1)} \equiv 1 \pmod{p}$
$\Rightarrow a^{q-1} \equiv 1 \pmod{p}$

④ $a^{q-1} \equiv 1 \pmod{p}$
[$\because p-1 \mid q-1$
$q-1 = k(p-1)$]

⑤ Using ② & ③,
$a^{q-1} \equiv 1 \pmod{p}$
$a^{q-1} \equiv 1 \pmod{q}$
$\therefore \boxed{a^{q-1} \equiv 1 \pmod{pq}}$ ($\because$ gcd $(p,q)=1$)
proved.

If p & q are distinct primes, then prove that: $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$

① Using F.L.T,
$a^{q-1} \equiv 1 \pmod{q}$
$a^{p-1} \equiv 1 \pmod{p}$

② $\therefore p^{q-1} \equiv 1 \pmod{q}$
$q^{p-1} \equiv 1 \pmod{p}$

③ Also, $q \equiv 0 \pmod{q}$
$\therefore q^{p-1} \equiv 0 \pmod{q}$

④ $\therefore$ from ② and ③,
$p^{q-1} + q^{p-1} \equiv 1 \pmod{q}$
Similarly,
$q\,p^{q-1} + q^{p-1} \equiv 1 \pmod{p}$

⑤ $\therefore \boxed{p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}}$
$\because$ gcd $(p,q)=1$

Let $n \equiv m \pmod{p-1}$
Then $a^n \equiv a^m \pmod{p}$ if $p \nmid a$

① Assume without loss of generality,
$n > m$
$n \equiv m \pmod{p-1}$
$n = c(p-1) + m$

② FLT $\Rightarrow a^{p-1} \equiv 1 \pmod{p}$
$\therefore a^{c(p-1)} \equiv 1 \pmod{p}$
$a^{n-m} \equiv 1 \pmod{p}$
$\therefore \boxed{a^n \equiv a^m \pmod{p}}$

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Generalisation of F·L·T

$$\gcd(a,m)=1 \Rightarrow a^{\varphi(m)} \equiv 1 \pmod{m}$$

Proof:

① Let $m=p^{\alpha}$ (i.e m is a power of prime)

② We prove using mathematical induction

③ For $\alpha=1$, $a^{\varphi(m)} = a^{\varphi(p)} = a^{p-1} \equiv 1 \pmod{p}$ [FLT]

④ Assume the result for some $\alpha-1$

$$a^{\varphi(m)} = a^{\varphi(p^{\alpha-1})} \equiv 1 \pmod{p^{\alpha-1}}$$

$$\Rightarrow a^{p^{\alpha-1}-p^{\alpha-2}} \equiv 1 \pmod{p^{\alpha-1}}$$

$$\Rightarrow a^{p^{\alpha-1}-p^{\alpha-2}} \equiv c \cdot p^{\alpha-1}+1 \quad \text{---} \circledast$$

⑤ raising both sides of $\circledast$ by p

$$(a^{p^{\alpha-1}-p^{\alpha-2}})^{p} \equiv (c \cdot p^{\alpha-1}+1)^{p}$$

$$a^{p^{\alpha}-p^{\alpha-1}} \equiv k \cdot p^{\alpha}+1$$

$$\therefore \quad a^{p^{\alpha}-p^{\alpha-1}} \equiv 1 \pmod{p^{\alpha}}$$

$$a^{\varphi(p^{\alpha})} \equiv 1 \pmod{p^{\alpha}}$$

⑤ Now, let $m=p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}$

$$a^{\varphi(p_1^{\alpha_1})} \equiv 1 \pmod{p_1^{\alpha_1}}$$

$$\left[a^{\varphi(p_1^{\alpha_1})}\right]^{\varphi(p_2^{\alpha_2})\cdots\varphi(p_r^{\alpha_r})} \equiv 1 \pmod{p_1^{\alpha_1}}$$

$$a^{\varphi(m)} \equiv 1 \pmod{p_1^{\alpha_1}} \quad (\because m=p_1^{\alpha_1}\cdots p_r^{\alpha_r} \text{ is multiplicative})$$

$\Rightarrow$ Similarly $a^{\varphi(m)} \equiv 1 \pmod{p_i^{\alpha_i}} \ \forall\ i \ 1 \le i \le r.$

⑥ $a^{\varphi(m)} \equiv 1 \pmod{p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_r^{\alpha_r}}$

$$\boxed{a^{\varphi(m)} \equiv 1 \pmod{m}}$$

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WILSON'S THEOREM

If p is a prime, then $(p-1)! \equiv -1 \pmod{p}$

Proof:

① For $p=2$, $(2-1)! \equiv 1 \equiv -1 \pmod 2$

$p=3$, $(3-1)! \equiv 2 \equiv -1 \pmod 3$

② For $p>3$, consider the congruences $ax \equiv 1 \pmod p$

for a, $1 \leq a \leq p-1$

$\therefore$ unique soln for all $a \in \{1, 2, \ldots, p-1\}$ as $p \nmid a$

③ $x \equiv a^{-1} \pmod p$ exists for all a $1 \leq a \leq p-1$

④ Let $a^{-1} = a$ for some a, $1 \leq a \leq p-1$

$a^{-1} \equiv a \pmod p$

$\Rightarrow a^2 \equiv 1 \pmod p$

$\Rightarrow p \mid a^2 - 1$

$p \mid (a+1)(a-1)$

$a-1 = 0$ & $a+1 = p$ [as $a+1 = p$ & $a-1 = p$ not poss]

$a = 1$ $a = p-1$ ($a=-1$ & $a = p+1$ not poss as $1 \leq a \leq p-1$)

(i.e $1 \times 1 \equiv 1$ & $(p-1)(p-1) \equiv 1$)

⑤ Each of $2, 3, 4, \ldots p-2$ have inverses mod p, which are distinct from themselves

$\Rightarrow (p-2)! \equiv 2 \times 3 \times \ldots (p-2)$

$\equiv 1 \pmod p$

(multiplying by $p-1$)

$(p-2)!(p-1) \equiv (p-1) \pmod p$

i.e $\boxed{(p-1)! \equiv -1 \pmod p}$

✱ Remark

$\Rightarrow (p-2)! \equiv 1 \pmod p$

✱ CONVERSE of Wilson Theorem is true

i.e n is prime, if

$(n-1)! \equiv -1 \pmod n$

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Theorem: The quadratic congruence $x^2+1 \equiv 0 \pmod p$ has a solution iff p is an odd prime such that $p \equiv 1 \pmod 4$

Proof:

① Let a be a soln of $x^2+1 \equiv 0 \pmod p$

$$a^2 \equiv -1 \pmod p \quad \text{—①}$$

② By F-L-T $a^{p-1} \equiv +1 \pmod p$

$$\Rightarrow (a^2)^{\frac{p-1}{2}} \equiv +1 \pmod p$$

$$\Rightarrow (-1)^{\frac{p-1}{2}} \equiv +1 \pmod p \text{ (from ①)}$$

③ If $p \equiv 3 \pmod 4$

$$\Rightarrow p = 4k+3$$

$$\Rightarrow \frac{p-1}{2} = \frac{4k+3-1}{2} = 2k+1$$

$\rightarrow \frac{p-1}{2}$ is odd

④ $\therefore (-1)^{\frac{p-1}{2}} \equiv -1 \pmod p$

& $-1 \equiv +1 \pmod p$

$\Rightarrow p \mid 2$ (not possible as $p > 2$)

⑤ $\therefore p \not\equiv 3 \pmod 4$

$\therefore p \equiv 1 \pmod 4$

⑥ Conversely, let $p \equiv 1 \pmod 4$

$p = 4k+1$

$\frac{p-1}{2} = 2k$ (even)

⑦ By Wilson Theorem,

$(p-1)! \equiv -1 \pmod p$

i.e $-1 \equiv 1 \times 2 \times 3 \times \dots (p-1) \pmod p$

$$\equiv 1 \times 2 \times \dots \left(\frac{p-1}{2}\right)\left(\frac{p+1}{2}\right)\dots(p-1) \pmod p$$

$$\equiv (1 \times -1).2(-2).3(-3) \times \dots \left(\frac{p-1}{2}\right)\left(-1\left(\frac{p-1}{2}\right)\right) \pmod p$$

$$-1 \equiv (-1)^{\frac{p-1}{2}} \left[1 \times 2 \times \dots \left(\frac{p-1}{2}\right)\right]^2$$

$$\equiv (-1)^{\frac{p-1}{2}} \left(\frac{p-1}{2}!\right)^2$$

⑧ But $(-1)^{\frac{p-1}{2}} \equiv +1$ $\left[\because \frac{p-1}{2} \text{ is even for } p \equiv 1 \pmod 4\right]$

$$\therefore \left[\left(\frac{p-1}{2}\right)!\right]^2 \equiv -1 \pmod p$$

$$x^2 \equiv -1 \pmod p$$

$\therefore$ ★ $x = \left(\frac{p-1}{2}\right)!$ is the solution to $x^2+1 \equiv 0 \pmod p$ iff $p \equiv 1 \pmod 4$

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Note : $(n-1)! \equiv 0 \pmod n$
if n is composite, except $n=4$

① For $n=4$, $(4-1)! = 3! = 6 \equiv 2 \pmod 4$

② Let $n > 4$
$\Rightarrow n$ is composite
$\Rightarrow n = r.s$
$\Rightarrow \gcd(n, n-1) = 1$
$\therefore 1 < r < n-1$
$\Rightarrow r$ is a factor in $(n-1)!$
Similarly, $1 < s < n-1$

③ If $r \neq s$, then r & s are distinct factors of $(n-1)!$
$\therefore n = r.s \mid (n-1)!$
$\Rightarrow (n-1)! \equiv 0 \pmod n$ —(i)

④ Let $r = s$,
$\Rightarrow n = r^2$
Now $r < \frac{n}{2}$
(as if $r \geq \frac{n}{2}$, then
$n = r^2 \geq \frac{n^2}{4}$
or $4r^2 \geq n^2$
$4n \geq n^2$
$4 \geq n$, but $n > 4$)
$\therefore r < \frac{n}{2}$
$\Rightarrow 2r < n$
or $2r \leq n-1$

⑤ $\Rightarrow r$ and $2r$ are two distinct factors of $(n-1)!$
$\therefore r(2r) \mid (n-1)!$
$r^2 \mid (n-1)!$
$n \mid (n-1)!$
$\Rightarrow (n-1)! \equiv 0 \pmod n$ —(ii)

⑥ Let p be an odd prime number, then
$(p-1)! \equiv (p-1) \pmod{1+2+\ldots+(p-1)}$

Proof: ① $(p-1)! \equiv (p-1) \pmod{\frac{p(p-1)}{2}}$
↑ To prove ↑

② $(p-1)! \equiv -1 \pmod p$ [Wilson's Theorem]
$(p-1)! \equiv p-1 \pmod p$ —(*)

③ $\frac{p-1}{2}$ is an integer, $\because p$ is odd

④ $\frac{p-1}{2} \mid (p-1)$ & ~~$p \mid (p-1)$~~
$p \mid (p-1)! - (p-1)$ (from ②)

⑤ $\therefore \frac{p-1}{2} \mid (p-1)! - (p-1)$

⑥ $\therefore$ from ④ & ⑤
$\frac{p-1}{2} \mid (p-1)! - (p-1)$ $\left[\because \frac{p-1}{2} < p-1\right]$
$\therefore (p-1)! \equiv (p-1) \bmod \left(\frac{p-1}{2}\right)$

⑦ $(p-1)! \equiv (p-1) \bmod \left(\frac{p.(p-1)}{2}\right)$ (*) $\left[\because \gcd\left(p, \frac{p-1}{2}\right) = 1\right]$

⑧ $(p-1)! \equiv (p-1) \bmod (1+2+\ldots+(p-1))$ ←

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(*) For any prime p & any integer a } prove that: ① $p \mid a^p + (p-1)!a$

② $p \mid (p-1)!a^p + a$

① FLT $\Rightarrow$ $a^p \equiv a \pmod p$

② Wilson's Theorem $\Rightarrow (p-1)! \equiv -1 \pmod p$

③ $a^p (p-1)! \equiv -a \pmod p$ [multiplying ① & ②]

$\Rightarrow \boxed{p \mid a^p (p-1)! + a}$

④ $a(p-1)! \equiv -a^p \pmod p$ [cross multiplying ① & ②]

$\Rightarrow \boxed{p \mid a(p-1)! + a^p}$

or

$(p-1)! a^p \equiv -a \pmod p$

$(p-1)! \frac{a}{a^{p-1}} \equiv -a \pmod p$

$(p-1)! a \equiv -a^p \pmod p$

$p \mid (p-1)! a + a^p$

(*) If p & $p+2$ are twin primes

Prove that: $4[(p-1)! + 1] + p \equiv 0 \pmod{p(p+2)}$

① Wilson's Theorem $(p-1)! \equiv -1 \pmod p$

$(p-1)! + 1 \equiv 0 \pmod p$

$4[(p-1)! + 1] \equiv 0 \pmod p$

$4[(p-1)! + 1] + p \equiv 0 \pmod p$ —— (*)

② Wilson's Theorem $((p+2)-1)! \equiv -1 \pmod{p+2}$ [$\because p+2$ is a prime]

~~$p+2$~~ $(p+1)! \equiv -1 \pmod{p+2}$

$(p+1)\, p! \equiv -1 \pmod{p+2}$

$(\cancel{p+1})\, p! \equiv (\cancel{p+1}) \pmod{p+2}$

$p! \equiv 1 \pmod{p+2}$

$4p! \equiv 4 \pmod{p+2}$

$4p! \equiv 4 + 2p - 2p \pmod{p+2}$

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$$4p(p-1)! \equiv 2(p+2) - 2p \pmod{p+2}$$

$$4p(p-1)! \equiv -2p \pmod{p+2}$$

$$4(p-1)! + 2 \equiv 0 \pmod{p+2}$$

$$4(p-1)! + 2 + p + 2 \equiv 0 \pmod{p+2}$$

$$4[(p-1)! + 1] + p \equiv 0 \pmod{p+2} \quad \text{——} (*)$$

③ From (*)'s,

$$4[(p-1)! + 1] + p \equiv 0 \pmod{p(p+2)}$$

$$[\because \gcd(p, p+2) = 1]$$

QUADRATIC RESIDUES

Let p be an odd prime ($p > 2$)

We say, $a \pmod p$ is a quadratic residue if $a \equiv b^2 \pmod p$

Otherwise, 'a' is a non-residue

Note: a is a residue mod p, if $a \equiv b^2 \bmod p$

For $b = 1, 2, 3 \ldots \frac{p-1}{2}$, $b^2 \equiv a \bmod p$

eg: 11 $p = 11$

$b = 1, 2 \ldots \frac{11-1}{2}$

$= 1, 2, 3, 4, 5$

$1^2 = 1 \pmod{11} \equiv 1$

$2^2 = 4 \pmod{11} \equiv 4$

$3^2 = 9 \pmod{11} \equiv 9$

$4^2 = 16 \pmod{11} \equiv 5$

$5^2 = 25 \pmod{11} \equiv 3$

$6^2 \equiv 36 \pmod{11} \equiv 3$

$7^2 = 49 \pmod{11} \equiv 5$

$8^2 = 64 \pmod{11} \equiv 9$

$9^2 = (-2)^2 \pmod{11} \equiv 4$

$10^2 = (-1)^2 \pmod{11} \equiv 1$

$\therefore$ 1, 3, 4, 5, 9 ⇒ quadratic residues

2, 6, 7, 8, 10 ⇒ non-residues

There are 50% residues & 50% non-residues

[Residues are squares of $1, 2, 3 \ldots \frac{p-1}{2}$]

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⊛ Generator mod p

Let p be a prime

Then, +ve integer g , $1 < g < p$ is a generator mod p

if $a = g^j \ \forall \ 1 \leq a \leq p-1$

i.e $(g^1, g^2, \ldots g^{p-1})$

g is a generator if it runs over complete set of residues (mod p)

⊛ $\{g, g^2, \ldots (g^{p-1} = 1)\} \equiv \{1, 2, \ldots p-1\} \mod p$

There will always be $\varphi(p-1)$ different generators (mod p)

If g is a generator, then g^j is also a generator iff

$\gcd(j, p-1) = 1$

eg : $p = 13 \Rightarrow$

$2^1 = 2$
$2^2 = 4$
$2^3 = 8$
$2^4 = 16 \equiv 3$
$2^5 = 32 \equiv 6$
$2^6 \equiv 64 \equiv -1$
$2^7 \equiv 11$
$2^8 \equiv 9$
$2^9 \equiv 5$
$2^{10} \equiv 10$
$2^{11} \equiv 7$
$2^{12} \equiv 1$

} powers of 2 run through all non-zero numbers (mod p)
$\Rightarrow$ 2 is a generator.

for $p = 13$, $\varphi(13-1) = \varphi(12) = 4$ generators

$2, 2^5, 2^7, 2^{11} \equiv 2, 6, 11, 7$ are 4 generators.

Note $\Rightarrow$ a (mod p) is a residue iff a is an even power generator.

(i.e If $a = g^j$ and j is even ($j = 2k$),

Then $a = g^{2k} = (\pm g^k)^2 \Rightarrow$ a is a residue

On the other hand, if $a = b^2$, then $(g^j) = (g^k)^2$

when $b = g^k$

$\therefore j = 2k$ (even)

if a is a quadratic residue, then a is a square of $\pm g^{j/2}$ where g is a generator mod p

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LEGENDRE SYMBOL

Let p be an odd prime. Then for any $a \in \mathbb{Z}$,

Legendre Symbol $\left(\frac{a}{p}\right)$ is:

$$\left(\frac{a}{p}\right) = \begin{cases} 0, & \text{if } p \mid a \\ 1, & \text{if } a \text{ is quad. residue } (\text{mod } p) \\ -1, & \text{if } a \text{ is non-residue} (\text{mod } p) \end{cases}$$

Theorem 1: $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$

Proof: ① If $p \mid a$, then $L.H.S = RHS = 0$

② Let $p \nmid a$. Then $a^{p-1} \equiv 1 \pmod{p}$ [by FLT]

$$(a^{\frac{p-1}{2}})^2 \equiv 1 \pmod{p}$$

$$a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$$

③ Let g be a generator $(\text{mod } p)$
& $a = g^j$

$\Rightarrow a$ is quadratic residue iff j is even

④ $a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$ [from ②]

$(g^j)^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$ [from ③]

$\therefore g^{j\frac{(p-1)}{2}} \equiv +1 \pmod{p}$ iff $(p-1) \mid j\frac{(p-1)}{2}$

$\Rightarrow j$ is even

$\left[\because g^{p-1} \equiv 1 \text{ as } g \text{ is generator}\right.$
$\left.\& \; g^{(p-1)k} \equiv 1 \text{ only if } k \in \mathbb{Z}\right]$

⑤ $\therefore a^{\frac{p-1}{2}} \pmod{p} \equiv \left(\frac{a}{p}\right)$

⊛ Properties

① $\left(\frac{a}{p}\right)$ is only dependent on $a \pmod{p}$

② $\left(\frac{ab}{}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$

③ If $\gcd(a, p) = 1$
Then $\left(\frac{ab^2}{p}\right) \equiv \left(\frac{a}{p}\right)$

④ $\left(\frac{1}{p}\right) = +1$ & $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$

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Theorem 2: $\left(\frac{2}{p}\right) \equiv (-1)^{\frac{p^2-1}{8}} = \begin{cases} +1 & \text{if } p \equiv \pm 1 \pmod 8 \\ -1 & \text{if } p \equiv \pm 3 \pmod 8 \end{cases}$

Law of Quadratic Reciprocity

Let p, q be odd primes

$$\left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}} \left(\frac{p}{q}\right) = \begin{cases} -(p/q) & \text{if } p \equiv q \equiv 3 \pmod 4 \\ +(p/q) & \text{otherwise} \end{cases}$$

(*)(*) SQUARE ROOT MODULO P. ($x^2 \equiv a \pmod p$).

① Find non-residue (mod p). ⇒

② Write (i) $p-1 = 2^\alpha . s$
(ii) $b = n^s$
(iii) $r = a^{\frac{s+1}{2}}$
} mod p

Find:
① a ④ s ⑦ n
② b ⑤ r ⑧ a^{-1}
③ α ⑥ p

③ Reqd: $x = b^j . r$ $\quad j = j_0 + 2j_1 + 2^2 j_2 + \cdots 2^{k-2} j_{k-2}$ $\quad k = \alpha - 2$

① To find $j_0 \Rightarrow (r^2 a^{-1})^{2^{\alpha-2}} \pmod p$ $\quad j_0 = \begin{cases} 0 & \text{if } 1 \\ 1 & \text{if } -1 \end{cases}$.

② To find $j_1 \Rightarrow ((b^{j_0} . r)^2 a^{-1})^{2^{\alpha-3}} \pmod p$ $\quad j_1 = \begin{cases} 0 & \text{if } 1 \\ 1 & \text{if } -1 \end{cases}$.

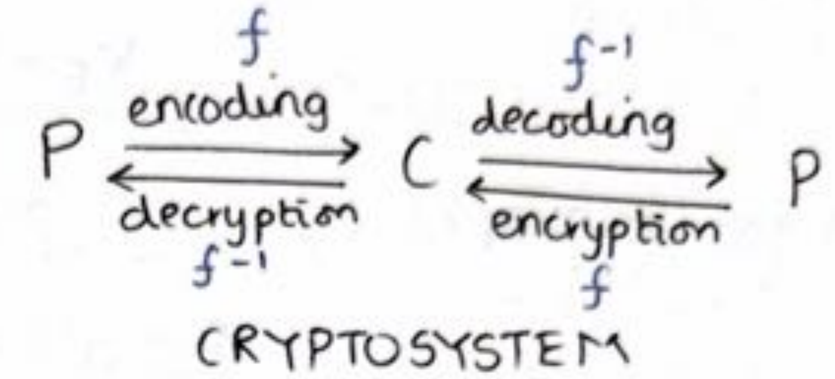
③ To find $j_2 \Rightarrow ((b^{j_0+2j_1} . r)^2 a^{-1})^{2^{\alpha-4}} \pmod p$ $\quad j_2 = \begin{cases} 0 & \text{if } 1 \\ 1 & \text{if } -1 \end{cases}$

⋮

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CRYPTOGRAPHY

Plaintext ⇒ messages in original form [P]
Ciphertext ⇒ messages in disguised form [C]



⊛ Message Units

k-graphs →
1. Single Letter Message Units ⇒ each alphabet is a message unit.
2. Diagraphs ⇒ pair of letters as message units — eg: COME → "CO", "ME"
3. Trigraphs ⇒ three letters together as a message unit — eg: YESHI ⇒ "YES", "HI_"

⊛ Labelling of plaintext / ciphertext.

- A-Z can be labelled as 0-25 [numerical equivalents]
- Special characters may be labelled as 26, 27, ... eg: blank = 26

for k-graph over N-letter alphabet,

$N^{k-1}x_1 + N^{k-2}x_2 + \dots + Nx_{k-1} + x_k$

$\in \{0, 1, 2 \dots, N^k - 1\}$

For digraph over N-letter alphabet numerical equivalent will be $Nx+y$

$\{x, y\}$ = (1st & 2nd letter of digraph) $= \{0, 1, 2 \dots 25^{N-1}\}$

$Nx+y \in \{0, 1, 2, \dots, N^2-1\}$

For trigraph over N-letter alphabet, numerical equivalent will be $N^2x + Ny + z$

$\in \{0, 1, 2, \dots N^3 - 1\}$

eg: YES ⇒ (26-letter) ⇒ $26^2(24) + 26(4) + 18$ (Y, E, S)

SHIFT TRANSFORM [CAESAR CIPHER]

$C \equiv P + b \pmod N$

$P \equiv C - b \pmod N$

P → plaintext
C → ciphertext
b → shift
N → alphabet.

eg: YES ⇒ P, b = 7

(Encryption) YES ⇒
- 24 + 7 ≡ 31 (mod 26) ⇒ 5 → F
- 4 + 7 ≡ 11 (mod 26) ⇒ 11 → L
- 18 + 7 ≡ 25 (mod 26) ⇒ 25 → Z

⇒ C = "FLZ"

(Decryption)
- F ⇒ 5 − 7 (mod 26) ⇒ −2 ≡ 24 → Y
- L ⇒ 11 − 7 (mod 26) ≡ +4 ≡ 4 → E
- Z ⇒ 25 − 7 (mod 26) ≡ 18 ≡ 18 → S

⇒ P = "YES"

AFFINE MAPPING

$C \equiv aP + b \pmod N$

$K_E = (a, b)$
↥ encryption key

Decryption:

$P \equiv a^{-1}(C - b) \pmod N$
$= a^{-1}C - a^{-1}b \pmod N$
$= a'C + b' \pmod N$

$P \equiv a'C + b' \pmod N$

$K_D = (a', b')$
$= (a^{-1}, -a^{-1}b)$

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Ex E ≡ $C \equiv aP + b \pmod N$ $\longleftrightarrow$ $P \equiv a^{-1}C - a^{-1}b \pmod N$

$K_E = (a, b)$ $\quad K_D = (a', b') = (a^{-1}, -a^{-1}b)$

(*) for k-graphs ⇒ $C \equiv aP + b \pmod{N^k}$

$P \equiv a'C + b' \pmod{N^k}$

eg: Ciphertext ⇒ "PWULPZTQAWHF"

Encryption ⇒ Affine Transf^ on Digraphs using 26 letter alphabet

Common Digraphs ⇒ IX ≡ TH Read the msg.

TQ ≡ HE

[C] [P]

① Write encryption scheme / formula

$C \equiv aP + b \pmod{N^2}$

$P \equiv a'C + b' \pmod{N^2}$ $\left[\begin{matrix} a' = a^{-1} \\ b' = -a^{-1}b \end{matrix}\right]$

② Use digraphs in the Q to get a, b / a', b'

[C] IX ⇒ $26 \times 8 + 23 = 231 \pmod{676}$ | TH ⇒ $26 \times 19 + 7 \equiv 501$ [P]

TQ ⇒ $26 \times 19 + 16 = 510 \pmod{676}$ | HE ⇒ $26 \times 7 + 4 = 186$

$P \equiv a'C + b'$

$501 \equiv a'(231) + b' \pmod{676}$ —①

$186 \equiv a'(510) + b' \pmod{676}$ —②

① − ② ⇒ $315 \equiv -279a' \pmod{676}$

$315 \equiv 397a' \pmod{676}$

$\therefore a' = 315 \times (397)^{-1} \pmod{676}$

$\equiv 315 \times 613 \pmod{676}$

$a' \equiv 435$

E·A

	676	~~01~~	9
1	397	0	1
1	279	1	-1
~~2~~ 2	2118	-1	+2
2	43	+3	~~-5~~
1	32	~~-7~~ -4	+12
2	11	+10	~~13~~ -17
1	10	-27	~~-35~~ 46
	~~3~~ 1	+37	~~48~~ -63

$\therefore b' \equiv 501 - 231a'$

$= 501 - 231(435)$

$\equiv -612 \pmod{676}$

~~$\equiv 64 \pmod{676}$~~ (don't take this as in next step, we have to subtract only)

$\therefore a' \equiv 435$

$b' \equiv -612$

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(3) Convert cipher digraph to numerical equivalent.

PW = 26 (15) + 22 = 412
UL = 26 (20) + 11 = 531
PZ = 26(15) + 25 = ~~510~~ 415
TQ = 26 (19) + 16 = ~~22.530~~ 510
AW = 26 (0) + 22 = 22
HF = 26 (7) + 5 = 187

(4) Use the formula to convert to plaintext

$P \equiv a'P + b' \pmod{676}$ (C)

PW = 412 $\xrightarrow{}$ 435(412) − 612 ≡ 178608
≡ 144
≡ 26(5) + 14
⇒ "FO"

HF = 187 → 435(187) − 612 ≡ 289
≡ 26 × ( ) + ( )
⇒ "ZD"

P = "FOUNDTHEGOLD"

ENCRYPTION/DECRYPTION OF DIGRAPH VECTORS

$C \equiv AP \pmod{N}$ i.e $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$ → A should be invertible

digraph vector → C, A, P

To decipher, we apply the inverse ⇒ $P \equiv A^{-1} C \pmod{N}$ i.e $\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{|ad-bc|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$

↳ $\frac{1}{D} \Rightarrow D^{-1}$ [calculate by finding GCD of (D, N)] (E·A)

⊛ To encipher plaintext sequence of k-digraphs $P_1 P_2 P_3 \ldots P_R$ ⇒

$$C = AP = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2\times 2} (P_1\ P_2\ P_3 \ldots P_k)_{2\times k}$$

$$= (C_1\ C_2\ C_3 \ldots C_k)$$

⊛ If we know certain P & C ⇒

$C \equiv AP$
$\therefore A = CP^{-1}$
$A^{-1} = PC^{-1}$

] to find A & A^{-1} } (mod N)

eg: "NOANSWER" ⇒ P & $A = \begin{pmatrix} 2 & 3 \\ 7 & 8 \end{pmatrix}$

P_1 = NO
P_2 = AN
P_3 = SW
P_4 = ER

$C = A_{2\times 2} P_{2\times k}$

$$= \begin{pmatrix} 2 & 3 \\ 7 & 8 \end{pmatrix} \begin{pmatrix} N & A & S & E \\ O & N & W & R \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 7 & 8 \end{pmatrix} \begin{pmatrix} 13 & 0 & 18 & 4 \\ 14 & 13 & 22 & 17 \end{pmatrix} \pmod{26}$$

$$\equiv \begin{pmatrix} 16 & 13 & 24 & 7 \\ 21 & 0 & 16 & 8 \end{pmatrix} = QVNAYQHI$$

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PUBLIC KEY CRYPTOGRAPHY

1) R·S·A

(Rivest, Shamir, Adleman) Cryptography.

→ Each user A selects two large primes p_A, q_A and one random number e_A which has no common factor with $\varphi(n) = \varphi(pq)$

→ • p_A, q_A
• $\varphi(n) = \varphi(p_A q_A) = (p_A - 1)(q_A - 1)$
$= (n_A + 1) - p_A - q_A$
• $\gcd(e_A, \varphi(n_A)) = 1$
$[1 < e_A < \varphi(n_A)]$

→ $d_A = e_A^{-1} \pmod{\varphi(n_A)}$ [computed using E·A]

① $K_E = (e_A, n_A)$ [public]
$K_D = (n_A, d_A)$ [private]
secret
② $n = \overbrace{pq}$
③ $\underbrace{\varphi(n)}_{\text{secret}} = (p-1)(q-1)$
$= (n+1) - p - q$
④ $\underbrace{d}_{\text{secret}} = e^{-1} \pmod{\varphi(n)}$

$\varphi(n_A)$ and d_A are kept secret ⇒ knowledge of $\varphi(n)$ is same as finding p & q, ∵ $n = pq$.

Encryption: $C \equiv f(P) \equiv P^{e_A} \pmod{n_A}$

Decryption: $P \equiv f^{-1}(C) = C^{d_A}$
$= (P^{e_A})^{d_A}$
$= P \pmod{n_A}$

⊛ $e^A d^A = 1 \pmod{\varphi(n)}$

#Note: PlainText → k-graph } $N^k < n < N^L$
CipherText → L-graph } N = no. of letters.

#Note: if $(a, m) = 1$
and $n \equiv n' \pmod{\varphi(m)}$
then $a^n \equiv a^{n'} \pmod{\varphi(m)}$

eg: Plaintext ⇒ "DONE"
Units ⇒ Digraph [P], Trigraph [C]
Encryption ⇒ RSA
N = 26-letter alphabet
public key = (899, 7)

① Convert digraph to numerical equivalent.
"DO" ⇒ 26×(3)+14 = 92
"NE" ⇒ 26×(13)+ 4 = 342

② Extract p, q from (n, e) ← public key K_E
① $n = 899 = 29 \times 31$
∴ $p = 29$
$q = 31$
② $\varphi(n) = \varphi(29 \times 31)$
$= (29-1)(31-1)$
$= 28 \times 30 = 840$

③ Calculate ciphertext.

③ $d = e^{-1} \pmod{\varphi(n)}$ ← calculate for C→P.
$d = 7^{-1} \pmod{840}$
In this case, d cannot be calculated ∵ $\gcd(7, 840) \neq 1$.

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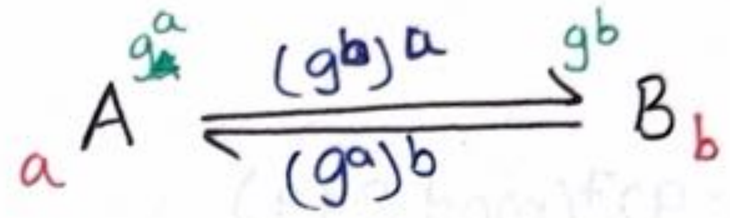
2) DIFFIE-HELLMAN KEY EXCHANGE SYSTEM

- A selects secret key $a \pmod p$
- B selects secret key $b \pmod p$

- A computes g^a and makes it public
- B computes g^b and makes it public

⇒ A & B exchange their common key $K = g^{ab}$

A computes g^{ab} using $(g^b)^a$
B computes g^{ab} using $(g^a)^b$.



⊛ g is a generator $(\text{mod } p)$
⊛ p → large prime
⊛ all computations are in $(\text{mod } p)$.

Q) enciphering transf^n $C \equiv AP + B \pmod N$
N = 31 - letter alphabet [blank = 26 ! = 29
. = 27 , = 30
? = 28]

You regard key $K_E = (A, B)$ as $A + Bi$ $(i^2 = -1)$

$g = 4 + i$

You choose $a = 209$
Your friend gives $g^b = 1 + 19i$

Encrypt
⇒ "BUVCFIWOUJJZ!H"

Given:
1) $a = 209$ ~~(mod 31)~~
2) $g^a = (4+i)^{209}$
3) $n = 31$
4) $g^b = (1+19i)$
5) msg units ⇒ single letters.
6) affine $C \equiv AP + B \pmod{31}$

① find shared key $K = g^{ab}$ $\pmod{31}$

$K = (g^b)^a = (1+19i)^{209}$
$(209 = 128 + 64 + 16 + 1)$
$(1+19i)^{209} = (1+19i)^{128+64+16+1}$
$= 18 \times 7 \times 14i\,(1+19i)$
$\equiv -33516 + 1764i$
$\equiv -5 + 28i$
$\equiv 26 + 28i$
$(A + Bi)$

$(1+19i)^1 \equiv (1+19i)$
$(1+19i)^2 \equiv 12+7i$
$(1+19i)^4 \equiv 2+13i$
$(1+19i)^8 \equiv 21+21i$
$(1+19i)^{16} \equiv 0+14i$
$(1+19i)^{32} \equiv -196 \equiv 21$
$(1+19i)^{64} \equiv 7$
$(1+19i)^{128} \equiv 18$

② Find secret keys ∴ $A = 26$, $B = 28$ $\pmod{31}$

③ find A^{-1} to find encryption

$A = 26 \pmod{31}$
$A^{-1} = 26^{-1} \pmod{31}$
$= 6 \pmod{31}$

[$26x \equiv 1 \pmod{31}$
$x = 6$
$26 \times 6 - 1 \equiv 0 \pmod{31}$]

④ find plaintext.

$C \equiv AP + B \pmod N$
$P = A^{-1}(C - B) \bmod N$
$P = 6(C - 28) \bmod 31$

eg: B → 1 $P = 6(1 - 28) \bmod 31 \Rightarrow 24 \Rightarrow Y$

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EL-GAMAL CRYPTOSYSTEM

[Discrete Logarithm Problem ⇒ given g and $g^x \pmod p$ computing x is hard
$x = \log_b y \pmod p$
$\Leftrightarrow y = b^x \pmod p$]

To send message P ⇒ choose random integer k and then send P in foll. pair (g^k, Pg^{ak})

① a ⇒ secret $(0 < a < p-1)$

② public key ⇒ (p, g, g^a) (p: prime, g: generator.)

③ ENCRYPTION: $C \equiv P(g^a)^k \equiv Pg^{ak}$ $[g^k, Pg^{ak}]$

④ DECRYPTION: $P \equiv (g^k)^{p-1-a}(C)$
$\equiv (g^k)^{p-1-a}(Pg^{ak})$
(p: prime; Pg^{ak}: PlainText.)

9) P = "REPLYTODAY"

$K_E = (47, 5, 10)$

k = 13

encrypt the msg.

$\Rightarrow K_E = (47, 5, 10)$
(p, g, g^a)

$(g^a)^k \pmod p$
$(10)^{13} \pmod{47}$

$10 \equiv 10 \pmod{47}$
$10^2 \equiv 53 \pmod{47}$
$10^4 \equiv 36 \pmod{47}$
$10^8 \equiv 27 \pmod{47}$

$10^{13} = 10^{8+4+1}$
$= 27 \times 36 \times 10$
$= 38$

R = 17	$\equiv 17(g^a)^k \equiv 17(10)^{13} \pmod p \equiv 35$
E = 4	$\equiv 4 \times 38 \equiv 11$
P = 15	$\equiv 15 \times 38 \equiv 6$
L = 11	$\equiv 11 \times 38 \equiv 42$
Y = 24	$\equiv 24 \times 38 \equiv 19$
T = 19	$\equiv 19 \times 38 \equiv 17$
O = 14	$\equiv 14 \times 38 \equiv 15$
D = 3	$\equiv 3 \times 38 \equiv 20$
A = 0	$\equiv 0 \times 38 \equiv 0$
Y = 24	$\equiv 24 \times 38 \equiv 19$

Decryption (secret key $a = 19$)
$\Rightarrow P \equiv (g^k)^{p-1-a}(Pg^{ak})$
$\equiv (38)^{47-1-19}(C)$
$(38,35) \equiv 38^{27}(35) \Rightarrow (17) \Rightarrow R$
$38^{27}(11) \Rightarrow (4) \Rightarrow E$

$C \equiv$	
(38,35)	(38,17)
(38,11)	(38,15)
(38,6)	(38,20)
(38,42)	(38,0)
(38,19)	(38,19)

$K_E = (47, 5, 10)$ (p g g^a)

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① Fermat's Factorization.

- $n = ab \iff n = x^2 - y^2$

- $n = x^2 - y^2 = (x+y)(x-y)$
  $\quad\quad\quad